CostVision
AI COST INTELLIGENCE

CostVision Implementation Blueprint

Secure enterprise deployment inside our network — and integration with CAPEE.
All CAD models, drawings and images stay inside the company. Verified in the code.

100% CAD stays internal | **CAPEE integration ready** | **4–6 weeks of changes** | **6-phase rollout**

Management briefing · July 2026 · Grounded in a line-by-line audit of the platform code

PURPOSE OF THIS MEETING

The decision we are asking for today

Approve Phases 1–2: the IT-Security assessment and the architecture design for deploying CostVision inside our network and connecting it to CAPEE. (~5–7 weeks, existing teams, no licence spend.)

Why now

The tool is built, tested (790 automated tests) and proven on live runs — the value is waiting on deployment, not development.

Why it is safe

CAD processing already runs fully inside the server. One configuration change closes the only external AI connection.

Why CAPEE wins

CAPEE keeps the workflow; CostVision adds the AI, physics and organisational memory behind it. No tool replacement.

BACKGROUND

What CostVision is — a quick recap

18 cost engines

Machining, casting, injection, PCB, software and more — physics-based, every figure traceable

CAD-to-Cost

Reads STEP/IGES models with a real geometry engine and auto-fills the costing

Photo-to-BOM

Costs a PCB from photographs — detects components and builds the bill of materials

Agentic AI

Learns from every analysis and real quote; suggests, self-corrects and raises findings on its own

Proven, not promised: estimating error cut from 10.9% to 0.3% after learning from 3 real quotes · £512k/yr of pricing issues found autonomously in the live demo · 790 automated tests protect it all.

TECHNOLOGY, SIMPLY EXPLAINED

What the tool is made of — in plain words

Frontend *What users see*	The web application in the browser — forms, dashboards, reports. Nothing is installed on laptops; it is served from our own server.
Backend *The engine room*	The server application that does the work: runs the 18 cost engines, the AI logic and all calculations. Runs on a standard company virtual machine.
Database *The memory*	Where rate libraries, saved costings and the AI knowledge base live. Standard corporate database (PostgreSQL), encrypted, backed up by IT.
CAD engine *The 3D model reader*	Specialist geometry software (OCCT — the same core used by CAD vendors) that measures the 3D model: volume, weight, walls, holes. Runs INSIDE the backend.
AI layer *The language brain*	The AI model used for vision and language tasks. Today it calls an external API — the ONE connection this blueprint moves inside our control.

SECURITY

The security requirement — and the key finding

IT-Security requirement: no CAD files, engineering drawings or images may leave the company network. Ever.

The key finding from the code audit: CAD files already never leave the server.

The geometry engine runs inside our backend and processes CAD models in memory — files are not even written to disk, let alone sent anywhere. The only outbound flows are the AI layer (photos and derived summaries) and two optional public feeds. One endpoint change closes the gap.

Why you can trust this: this is not a vendor claim — we audited every outbound network call in the platform's source code, line by line, and listed each one. The full inventory is in the written plan.

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DATA-FLOW MAP

Where data flows today — verified in the code

STAYS INSIDE — already ✓

CAD / geometry processing
OCCT engine runs in the backend; files processed in memory, never stored

All 18 cost engines
Physics + maths, fully local

Knowledge base & learning loop
Similarity, calibration, autonomous findings — local database

BOM file parsing, exports, reports
Local parsers, local PDF/Excel generation

LEAVES TODAY — to fix

AI layer calls
Part/PCB photos + CAD-derived summaries → route to a private AI endpoint (§ next slides)

Live part pricing (optional, OFF by default)
Part-number text only — keep off, or approve separately

News ticker feeds
Public news fetch, no company data — switch off or proxy

DECISION

Deployment options — our recommendation

OPTION A

Fully air-gapped

Everything on our VMs. AI features off (or a self-hosted model later). Deterministic engines fully working.

+ Zero external connections — provable today

– Reduced AI capability

OPTION B ★ RECOMMENDED

Private AI, on-prem core

Platform on our VMs. AI calls go to Claude running in OUR OWN cloud tenancy over a private link — no public internet, no data retention.

+ Full capability + contractual & technical data control

– Requires cloud tenancy sign-off

OPTION C

Public AI API

Platform on-prem but AI calls to the public API.

+ Simplest

– Fails our requirement — photos would traverse a public API

Recommendation: Option B — with Option A available immediately for the most sensitive programmes.

ARCHITECTURE

Target architecture — everything inside our walls

COMPANY INTERNAL NETWORK

Engineers (browser) + CAPEE
single sign-on (Azure AD)

Corporate API Gateway
authentication · rate limits · audit logs

CostVision server (frontend + backend on our VM)
geometry engine (CAD, in-memory) · 18 cost engines · learning loop · report generation

Corporate database (PostgreSQL)
rates · knowledge base · encrypted at rest

Internal AI gateway
ONLY allowed exit · inspected · logged

Our cloud tenancy
Claude on AWS Bedrock / Google Vertex
private link · no public internet · zero data retention

Zero-trust rules

- Every request authenticated (Azure AD)
- Deny-all egress except the AI gateway
- Role-based access (engineer / lead / admin / CAPEE service)
- CAD pulled from PLM vault, never stored

CAD DATA PROTECTION

How CAD data is protected

CAD model opened & measured	Inside our backend (OCCT engine)	✓ **Already local — verified**
Feature detection (holes, threads, walls)	Inside our backend	✓ **Already local — verified**
BOM extraction from files	Inside our backend (local parsers)	✓ **Already local — verified**
Cost calculation & learning	Inside our backend + our database	✓ **Already local — verified**
Photo analysis & AI narrative	Private AI endpoint (Option B)	→ **The one change we make**
CAD file storage	Nowhere — processed in memory only	✓ **Files never written to disk**

Plus: logs never contain CAD payloads; metadata goes to the corporate SIEM; an "air-gapped" switch lets security PROVE zero egress in a witnessed firewall test.

INTEGRATION

CAPEE + CostVision — better together

CAPEE (stays the front door)

- Costing workflow, approvals, reporting
- System of record — unchanged for users
- Calls CostVision services behind the scenes

requests

AI answers

CostVision (the engine behind it)

- 18 physics cost engines + CAD reading
- AI memory: similar parts, self-calibration
- Autonomous findings for sourcing

What flows over the connection (internal APIs that already exist)

Cost a part → full should-cost with breakdown

CAD file → geometry + auto-filled inputs

New part → similar past parts + suggestions

PO price → feeds the learning loop automatically

Dashboard ← autonomous findings (£/yr)

One shared → rate library & knowledge database

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SCOPE OF WORK

What each side needs to change

CostVision — 6 bounded items

1. **Private AI routing —** point the AI client at our endpoint (config + small factory)
2. **"Air-gapped" switch —** provably disable all external calls
3. **Azure AD sign-on —** replace local logins with company SSO
4. **PostgreSQL —** move from file database to the corporate DB estate
5. **Cost API wrapper —** one clean endpoint per commodity for CAPEE
6. **Audit hardening —** admin audit table + logs to SIEM

CAPEE — small, additive changes

Backend · Small

HTTP client to call CostVision + token handling; hook PO prices into the learning API

User interface · Medium

an "AI insights" panel on the costing screen; a CAD-upload button; a findings widget

Data · Small–Medium

adopt the shared rate library; one-off import of historical quotes to seed the memory

Total critical path: ~4–6 engineering weeks. No architectural surgery on either side.

COMPLIANCE

Security & compliance — how we tick the boxes

Encryption

TLS everywhere in transit; encrypted database at rest; CAD never stored at all

Access control

Azure AD single sign-on; roles mapped to AD groups; CAPEE uses a least-privilege service account

Data-loss prevention

One AI exit door, inspected and logged; firewall denies everything else; witnessed air-gap test

Audit trails

Every rate change, knowledge write and admin action attributable to a user; logs to corporate SIEM

Standards mapping: ISO 27001 — covered by the controls above, add to ISMS scope · SOC 2 — inherited from the cloud tenancy (Bedrock/Vertex are certified) · GDPR — minimal personal data, zero-retention AI, EU region · ISO/SAE 21434 — engineering IT tool, out of vehicle scope; supply-chain review (SBOM) in CI.

PLAN

Rollout — six phases, value from week eight

1 · Security assessment 2–3 wks	**2 · Architecture design** 3–4 wks	**3 · Pilot (dummy CAD)** 2 wks	**4 · CAPEE pilot** 4–6 wks	**5 · Enterprise rollout** 4 wks	**6 · Monitor & govern** ongoing

Phase 3 exit gate
Security-witnessed test: firewall blocks all egress, full test suite passes, zero unexpected traffic.

Phase 4 exit gate
A cost engineer completes a real costing from inside CAPEE with AI insights; accuracy dashboard live.

First value
Week ~8: pilot users get similar-part suggestions; sourcing sees the first autonomous findings.

HONEST VIEW

Risks — and how we manage them

Cloud tenancy not approved

Likelihood: Medium

Mitigation: Fall back to Option A (air-gapped) — deterministic engines keep full value today; add a self-hosted vision model later if needed.

Learning depends on logged quotes

Likelihood: Medium

Mitigation: The CAPEE PO-price hook automates it — quotes flow in without anyone changing habits. Plus a one-off historical import to start smart.

Single-team knowledge of the platform

Likelihood: Low-Med

Mitigation: 790 automated tests, written architecture docs, and a named CAPEE-side maintainer trained during Phase 4.

Adoption ("another tool")

Likelihood: Low

Mitigation: Users stay in CAPEE — CostVision works behind the scenes. Nothing new to learn except better answers appearing.

DECISION

Feasibility verdict — and the ask

Secure deployment, CAD fully internal	**FEASIBLE — CAD already never leaves; one AI endpoint change closes the gap**
CAPEE + CostVision integration	**FEASIBLE — API-first design; CAPEE stays the front door**
Changes required	**Bounded: ~4–6 engineering weeks critical path**
Long-term governance	**Rate-library board · monthly findings review · quarterly access & egress audit**

The ask today: approve Phase 1 (IT-Security assessment) and Phase 2 (architecture design) — 5–7 weeks, existing teams. The full written plan and security checklist are ready for the security review.